



IT9aL – PROFESSIONAL TRACK FOR IT 3

Helpdesk Ticketing System (AidDesk)

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1. Project Overview

The traditional process of handling customer support and technical issues in organizations is often manual, inconsistent, and inefficient, which does not align with today's fast-paced technological environment. Managing support requests through verbal communication, paper logs, or scattered messaging platforms makes it difficult to track issues, assign responsibilities, and monitor resolution progress, leading to delays, miscommunication, and poor service quality due to limited personnel and lack of centralized management. In many organizations, support staff struggle to organize multiple requests simultaneously, especially when there is no structured system to categorize, prioritize, and assign tickets, while requesters also experience inconvenience as they cannot easily track the status of their concerns or receive timely updates, resulting in reduced productivity and dissatisfaction among both support teams and users [1].

A Helpdesk Ticketing System is a practical and modern solution that addresses these challenges by providing a centralized platform for managing support requests. It enables organizations to automate ticket creation, classification, prioritization, and assignment, leading to improved response times and better resource allocation [2]. Furthermore, a web-based ticketing system enhances communication between requesters and support staff, ensuring that issues are resolved in an organized and transparent manner [3]. This approach not only improves operational efficiency but also enhances user satisfaction by making support services more accessible, reliable, and efficient.

The system will be developed using the Laravel MVC framework:

- **Model** handles business logic and database interaction
- **View** handles the presentation layer and user interface
- **Controller** processes requests and coordinates models and views

1.1 Existing Systems

A. Freshdesk

Freshdesk is a cloud-based helpdesk ticketing system aimed at small to midsize businesses. It centralizes incoming support requests from multiple channels into a single platform, enabling agents to efficiently track, categorize, and respond to tickets [4].

Pros:

- Free trial available
- Automated time logging for tickets
- Customizable dashboards with visual statistics
- User-friendly interface
- Supports multiple email accounts

Cons:

- Limited automation for ticket categorization; manual editing required
- No separate resolution template for tickets; replies must be sent manually

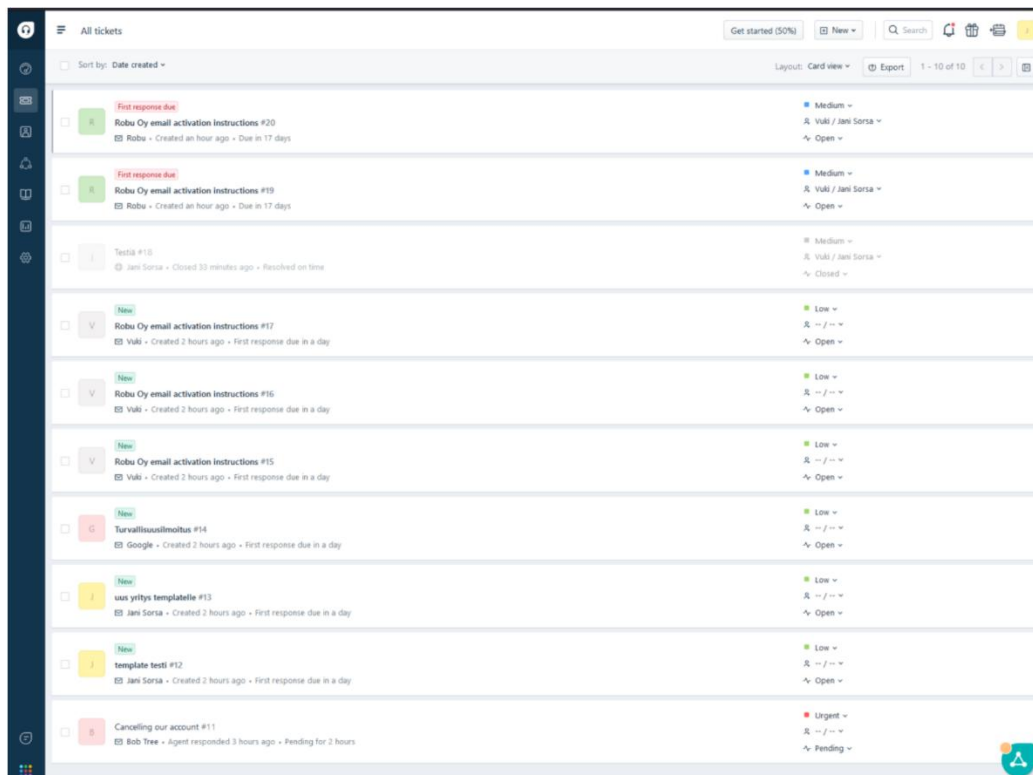


Figure 1: Freshdesk

B. Zoho Desk

Zoho Desk is a scalable SaaS helpdesk system designed for small to midsize businesses, capable of supporting growth to enterprise-level operations. It offers ticket automation, department-specific reporting, and accurate time tracking for agent productivity [5].

Pros:

- Efficient ticket automation and default templates for incoming emails
- Accurate time tracking for agents
- Ability to provide resolutions to customers
- Competitive pricing
- Supports department-based ticket organization

Cons:

- Reports are department-specific; manual compilation required for overall statistics

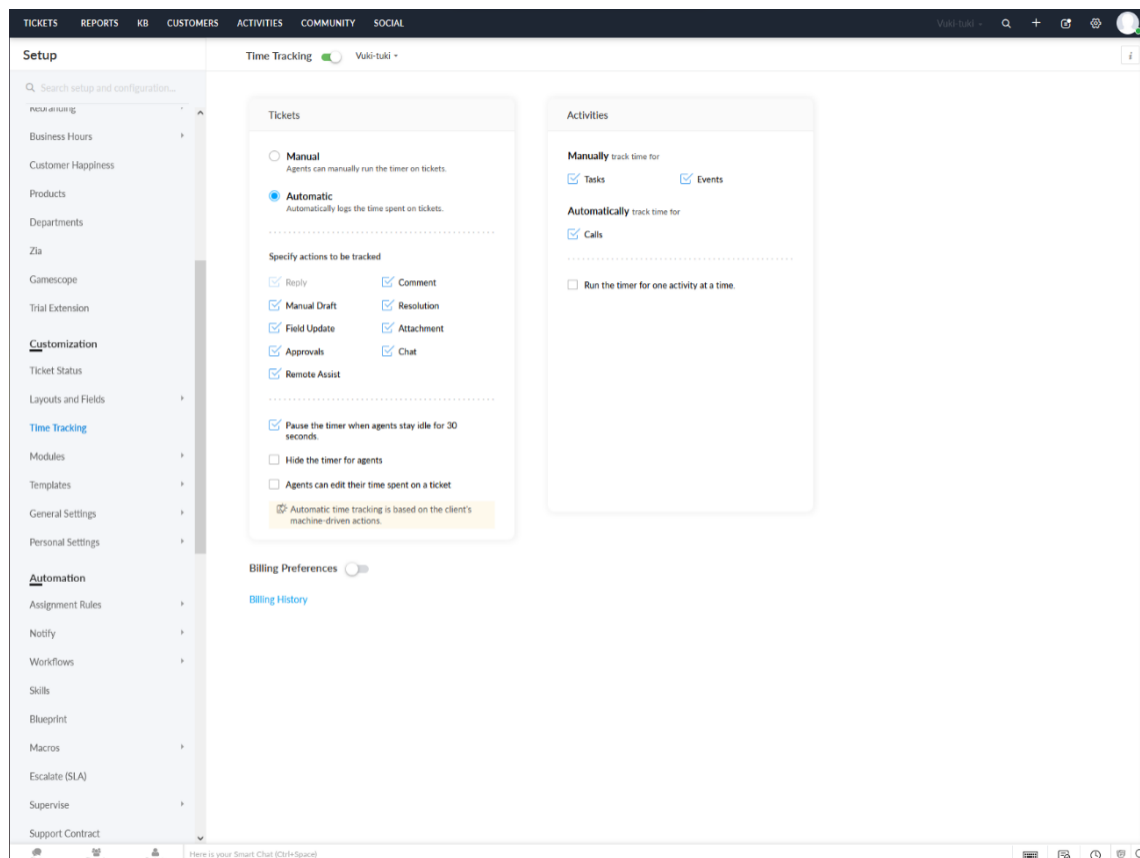


Figure 2: Zoho Desk

C. Bebek Kaleyo Ticketing System

A prototype-based IT helpdesk ticketing system for reporting, monitoring, and resolving IT equipment issues across multiple restaurant branches [6].

Pros:

- Speeds up reporting of IT problems compared to manual calls.
- Allows users to track ticket status in real-time via the dashboard.
- Improves IT department efficiency and job allocation through the “Assignment” feature.
- Reduces work interruptions by enabling faster issue resolution.
- User-friendly interface and accessible for branch staff.
- Supports better communication between users and IT staff.
- Provides a central digital record of all tickets.

Cons:

- Prototype design may still need refinement to enhance user-friendliness.
- Some features could be incomplete or require further enhancement based on user feedback.
- Limited to IT-related issues; may not support other business processes.
- Relies on all users being registered and logged in to access the system.

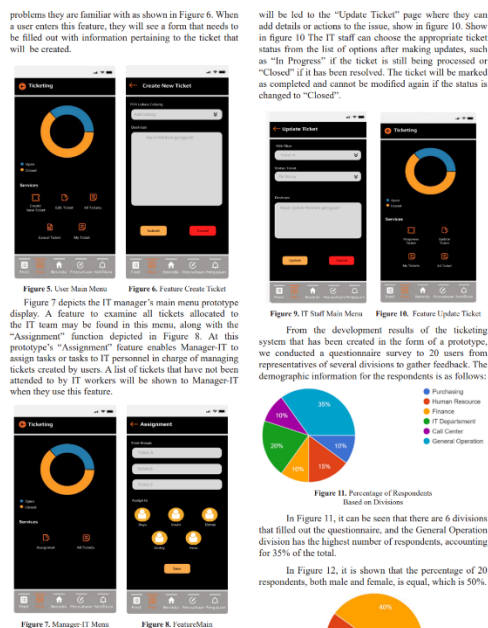


Figure 3: Bebek Kaleyo Ticketing System

2. Objectives

General Objective

To develop a Laravel-based Helpdesk / Ticketing System that organizes support requests, ticket assignments, issue updates, and resolution records efficiently. This study will develop a Helpdesk Ticketing System (AidDesk) with the following objectives:

1. To implement a requester management module that stores and maintains accurate customer or user information for efficient tracking of support requests.
2. To design a ticket management module that enables users to create, update, and monitor support tickets, including setting priority levels and tracking status in real time.
3. To develop a ticket assignment feature that allows proper distribution of tasks to support agents, ensuring accountability and faster resolution of issues.
4. To generate reports that provide insights into ticket status, agent performance, and overall support operations.
5. To maintain organized digital records of tickets, replies, and system activities for easy tracking, monitoring, and retrieval of past support transactions.

3. Scope

In Scope

- User authentication and authorization
- Requester management
- Ticket management
- Ticket assignment
- Status and priority tracking
- Ticket comments and update logging
- Dashboard and reporting
- Role-based access control

Limitations

The system is limited to web-based access and requires a stable internet connection for proper functionality. It does not include advanced features such as AI-based ticket classification or real-time chat support. Additionally, the system is designed primarily for small to medium-sized organizations and may require further optimization for large-scale deployment.

4. Users and Roles

Admin

- Manage all users
- Manage ticket categories and settings
- View all tickets and reports
- Configure system access

Support Agent

- View assigned tickets
- Update ticket status
- Add replies and resolution notes
- Manage ticket progress

Supervisor

- Monitor team performance
- Reassign tickets
- Review escalated issues
- View support reports

Requester

- Submit support tickets
- View ticket history
- Add follow-up replies
- Track ticket status

5. Functional Requirements

5.1 Authentication Module

Description: Handles login, logout, profile management, and role-based access.

Features:

- User login
- User logout
- Password hashing
- Role-based middleware
- Profile management

Laravel MVC Mapping:

- **Model:** User
- **Controller:** AuthController or Laravel built-in authentication controllers
- **View:** Login, Register, and Profile pages

5.2 Dashboard Module

Description: Displays ticket summaries and recent support activity.

Features:

- Total open tickets
- Total resolved tickets
- Tickets assigned to the current agent
- High-priority tickets
- Recent ticket updates

Laravel MVC Mapping:

- **Model:** Ticket, TicketReply, Category
- **Controller:** DashboardController
- **View:** Dashboard page

5.3 Requester Management Module

Description: Stores and manages requester or customer records.

Features:

- Add requester

- Edit requester
- Delete requester
- View requester details
- Search and filter requester records

Fields:

- Requester ID
- First Name
- Last Name
- Email
- Phone Number
- Department or Company
- Created At

Laravel MVC Mapping:

- **Model:** Requester
- **Controller:** RequesterController
- **View:** Requester list, create, edit, and details pages

5.4 Ticket Management Module

Description: Handles ticket creation, assignment, tracking, and resolution.

Features:

- Create ticket
- Edit ticket
- Assign ticket to agent
- Update ticket status
- Set priority
- Add issue description
- Close or reopen ticket

Ticket Status:

- Open

- In Progress
- Pending
- Resolved
- Closed

Fields:

- Ticket ID
- Requester ID
- Category ID
- Subject
- Description
- Priority
- Status
- Assigned User
- Created By
- Created At

Laravel MVC Mapping:

- **Model:** Ticket
- **Controller:** TicketController
- **View:** Ticket list, create, edit, and details pages

5.5 Ticket Reply / Update Log Module

Description: Records communication and updates related to each ticket.

Features:

- Add ticket reply
- Add internal note
- View ticket conversation history
- Track status changes and updates

Fields:

- Reply ID
- Ticket ID
- User ID
- Message
- Reply Type
- Created At

Laravel MVC Mapping:

- **Model:** TicketReply
- **Controller:** TicketReplyController
- **View:** Reply form and ticket conversation page

5.6 Category Management Module

Description: Organizes tickets by support category.

Features:

- Add category
- Edit category
- Delete category
- View category list

Fields:

- Category ID
- Name
- Description
- Status

Laravel MVC Mapping:

- **Model:** Category
- **Controller:** CategoryController
- **View:** Category list and form pages

5.7 Reports Module

Description: Generates summaries for support operations and staff monitoring.

Features:

- Open tickets report
- Resolved tickets report
- Agent performance report
- Ticket priority report
- Category-based ticket report

Laravel MVC Mapping:

- **Model:** Ticket, TicketReply, Category
- **Controller:** ReportController
- **View:** Reports dashboard

5.8 System Workflow

The system follows a structured workflow that ensures efficient handling of support requests from submission to resolution. The process begins when a requester submits a ticket containing relevant details such as subject, description, category, and priority level. The system records the ticket and assigns it an initial status of “Open,” after which it may be assigned to a support agent either manually or by an Supervisor. The assigned agent reviews the concern, updates the ticket status to “In Progress,” and communicates with the requester through replies when necessary. Supervisors may monitor or reassign tickets to ensure proper handling. Once the issue has been addressed, the agent marks the ticket as “Resolved,” and upon confirmation or completion, the ticket is finally set to “Closed.” This workflow ensures proper tracking, accountability, and timely resolution of reported issues.

6. Non-Functional Requirements

- The system must be accessible through a web browser
- The interface must be responsive

- Passwords must be securely hashed
- Only authorized users may access protected modules
- Data must be stored in a relational database
- The system should support CRUD operations efficiently
- The application should follow Laravel coding standards

6.1 Security Requirements

Security is a critical component of the AidDesk Helpdesk Ticketing System to ensure the protection of sensitive user and organizational data [7]. The system implements authentication mechanisms to verify user identity and uses role-based access control to restrict system functionalities according to user roles. Passwords are securely stored using hashing techniques to prevent unauthorized access. Input validation and data sanitization are applied to reduce the risk of common vulnerabilities such as SQL injection and cross-site scripting. In addition, built-in security features of the Laravel framework, including cross-site request forgery protection, are utilized to safeguard user interactions. These measures collectively help maintain data integrity, confidentiality, and system reliability [8].

7. Database Tables

users

- id
- name
- email
- password
- role
- created_at
- updated_at

requesters

- id
- first_name

- last_name
- email
- phone
- department
- created_at
- updated_at

categories

- id
- name
- description
- status
- created_at
- updated_at

tickets

- id
- requester_id
- category_id
- subject
- description
- priority
- status
- assigned_user_id
- created_by
- created_at
- updated_at

ticket_replies

- id
- ticket_id

- user_id
- message
- reply_type
- created_at
- updated_at

8. Entity Relationships

- One requester can create many tickets
- One category can contain many tickets
- One support agent can be assigned many tickets
- One ticket can have many replies
- One user can create many ticket replies
- One user can create many tickets

8.1 ERD

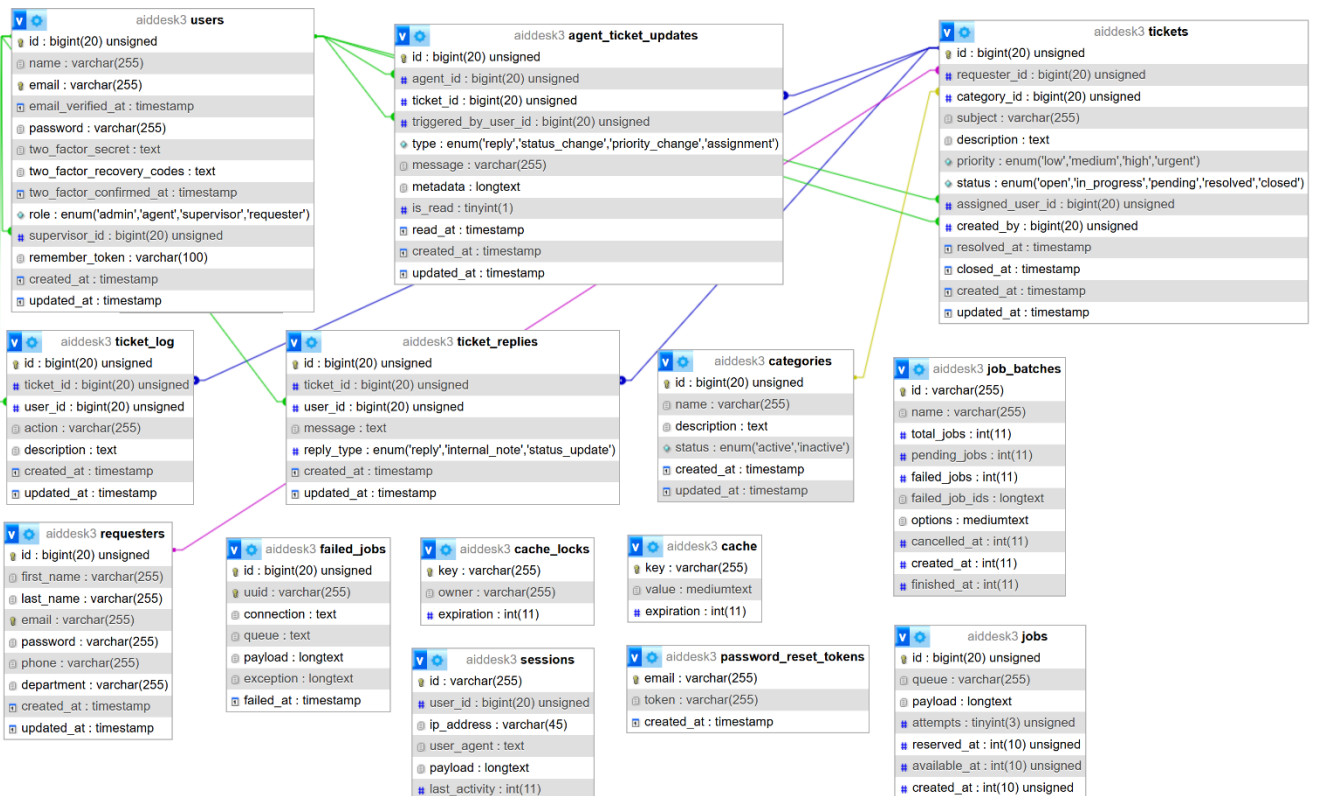


Figure 4: AidDesk ERD

9. Business Rules

- Only authenticated users can access the system
- Only admins can manage user accounts
- Every ticket must belong to one requester
- Every ticket must have a valid category, priority, and status
- A ticket may be assigned to only one support agent at a time
- Closed tickets cannot be edited unless reopened
- Every ticket reply must be linked to a user
- Reports must only display data allowed by user role

10. Validation Rules

Requester

- First name: required
- Last name: required
- Email: required, valid, unique
- Phone: required

Ticket

- Requester ID: required
- Category ID: required
- Subject: required
- Description: required
- Priority: required
- Status: required

Ticket Reply

- Ticket ID: required
- Message: required
- Reply type: required

11. Views

- Login page
- Dashboard
- Requester list
- Add requester form
- Edit requester form
- Ticket list
- Ticket details page
- Ticket reply page
- Category list
- Reports page

11.1 UI Design

This section presents the user interface (UI) design of the system, highlighting how the design integrates the following elements to ensure usability and functionality. The design integrates the following components:

- Role-based dashboards (Admin, Support Agent, Supervisor, Requester)
- Real-time ticket updates and notification feeds
- Comprehensive ticket management workflow
- Advanced reporting and analytics system
- User and category management tools
- Glassmorphism UI with neutral/black theme
- Priority and status indicators
- Response time tracking and SLA monitoring
- Enhanced search and filtering capabilities
- Audit trail and activity logging

Admin:

- Complete system oversight and configuration
- User management (create, edit, assign roles)

- Category and department management
- System-wide reports and analytics
- Agent performance monitoring
- Settings and customization controls

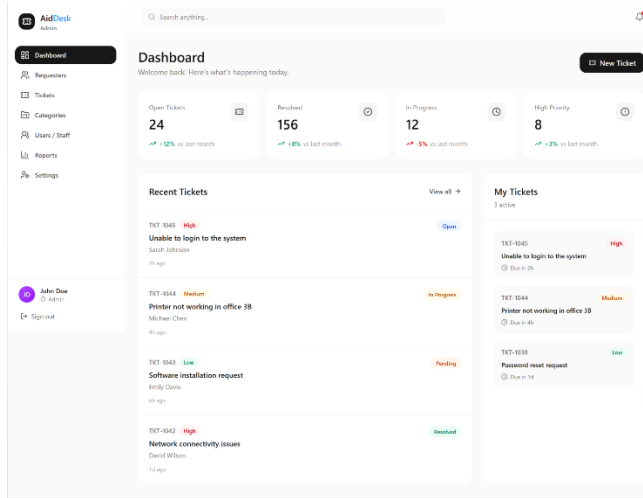


Figure 6: Admin_Dashboard

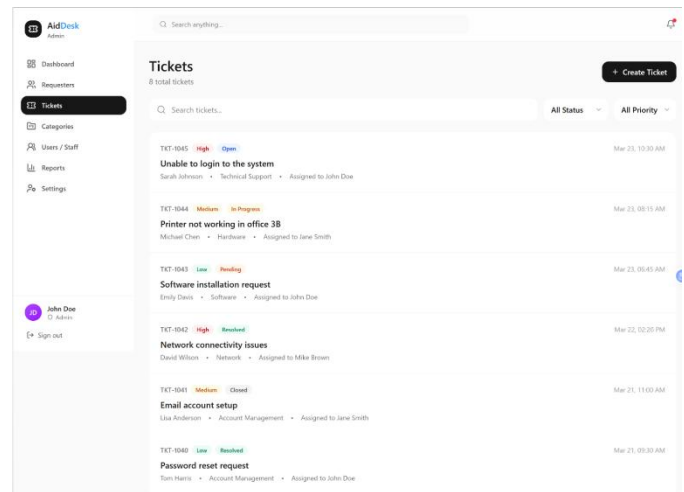


Figure 5: Admin_Tickets

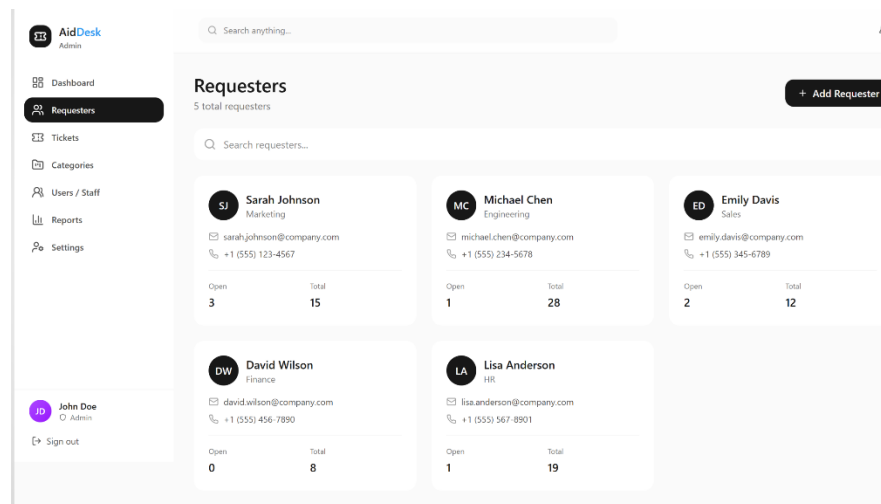


Figure 7: Admin_Requester

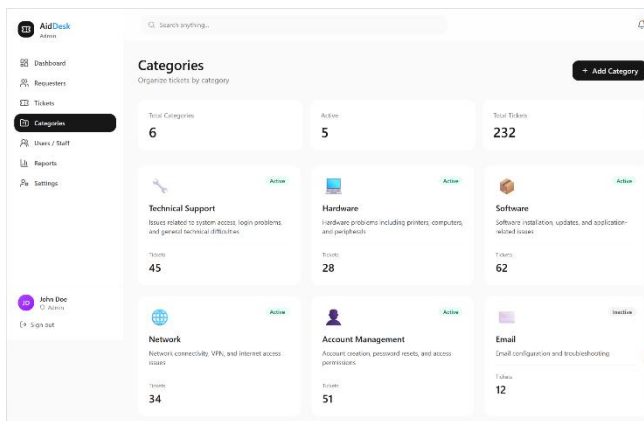


Figure 9: Admin_Categories

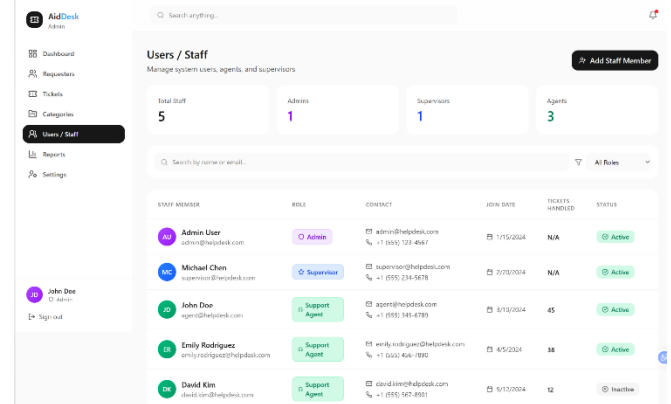


Figure 8: Admin_Users

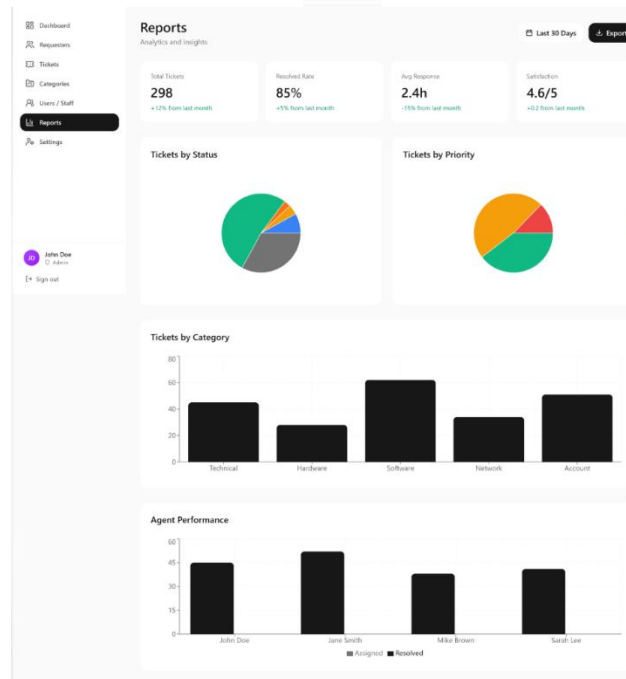


Figure 10: Admin_Reports

Support Agent:

- Assigned ticket queue and management
- Ticket Updates notification feed with unread indicators
- Ticket response and resolution tools
- Status and priority updates
- Activity logging and notes
- Personal performance dashboard

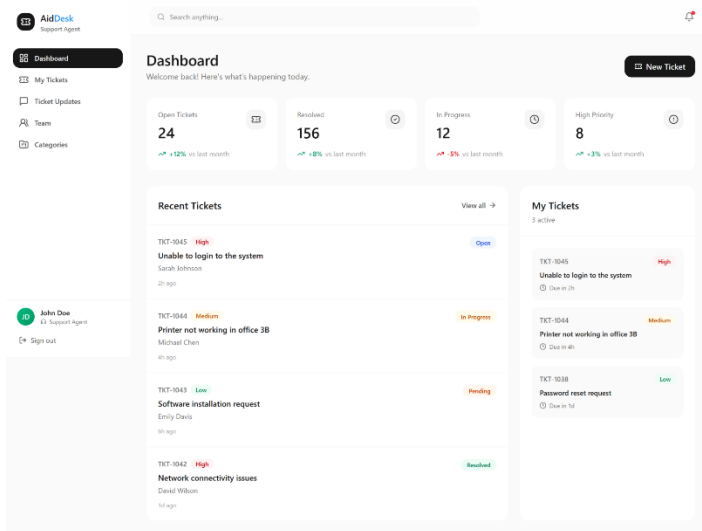


Figure 11: Agent_dashboard

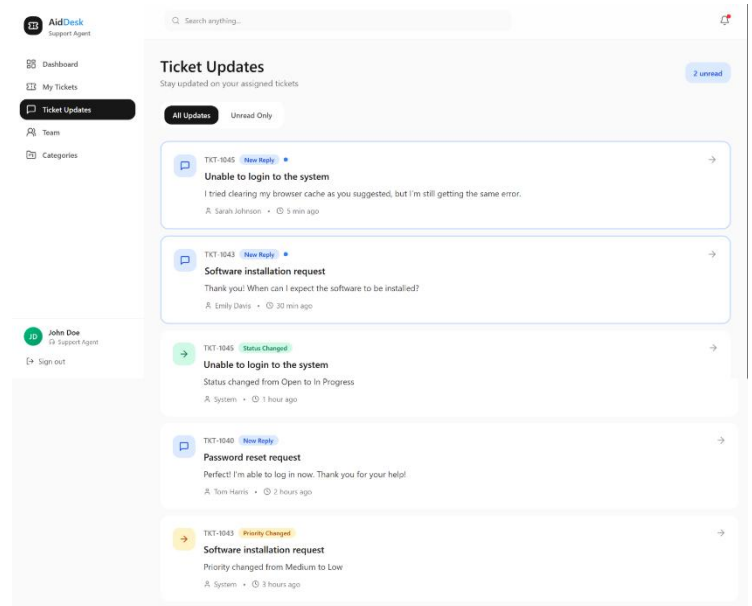


Figure 12: Agent_notification

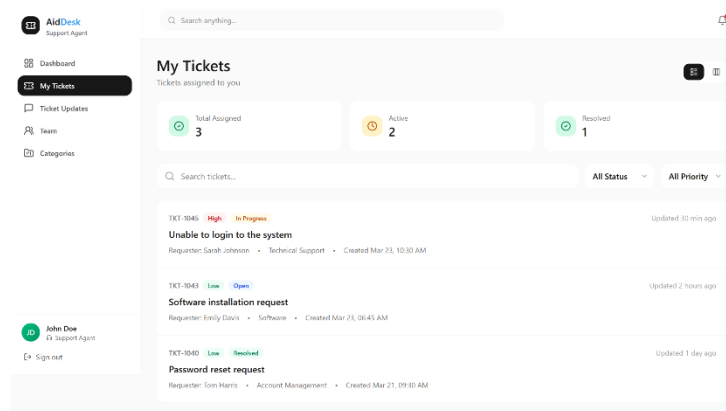


Figure 13: Agent_Tickets

Supervisor:

- Team oversight and ticket assignment
- Agent performance reports
- Workload distribution monitoring
- Escalation management
- Department-level analytics
- Quality assurance tools

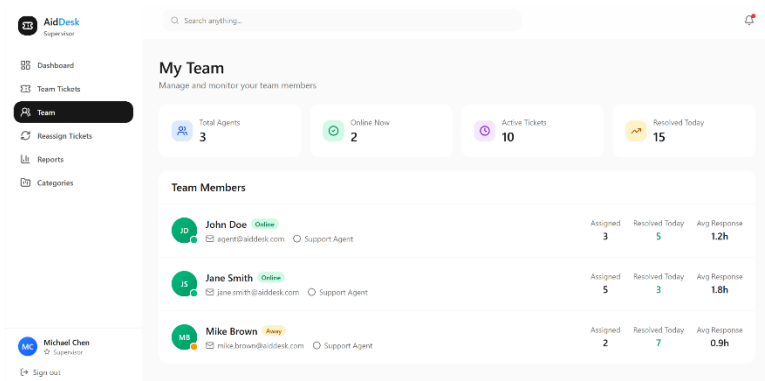


Figure 14: Supervisor_team_section

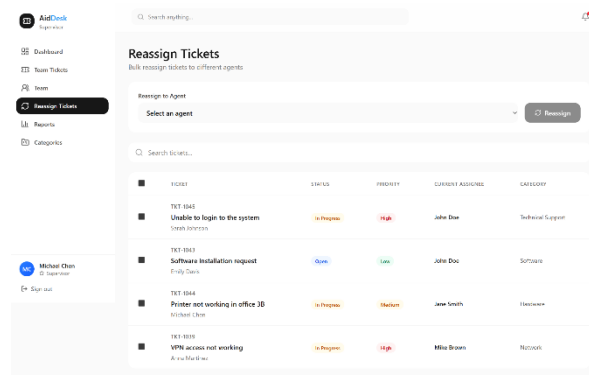


Figure 15: Supervisor_Tickets_Reassignment

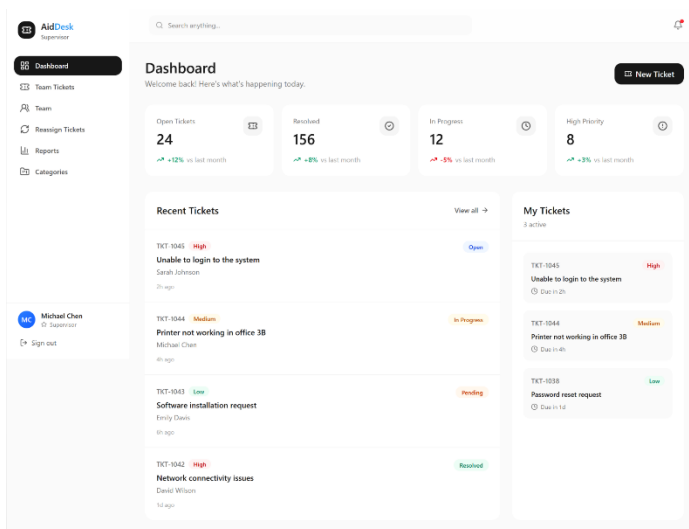


Figure 16: Supervisor_dashboard

Requester:

- Ticket creation interface
- My Tickets tracking dashboard
- Ticket status monitoring
- Response viewing and communication
- Ticket history and updates
- Simple, customer-focused UI

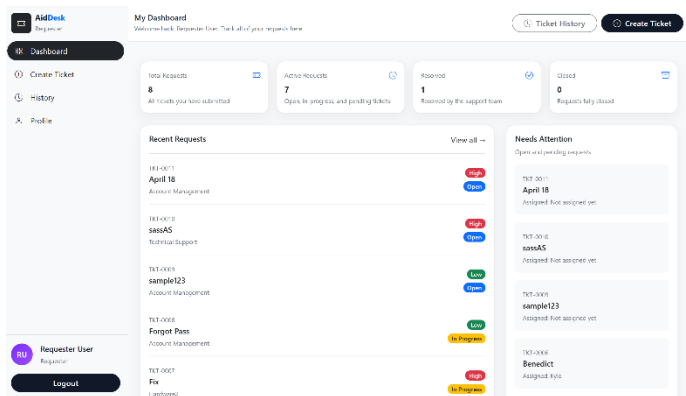


Figure 17: Requester_dashboard

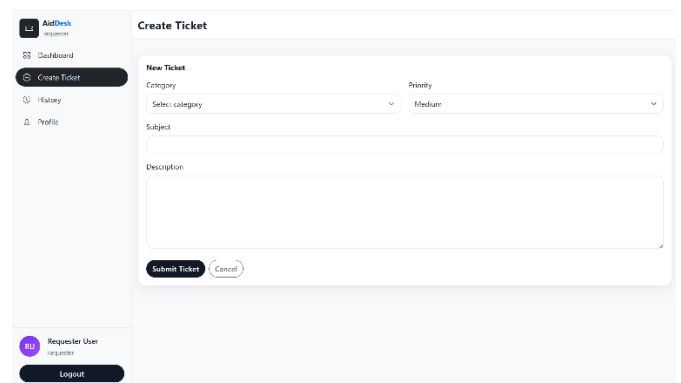


Figure 18: Requester_Tickets_Request

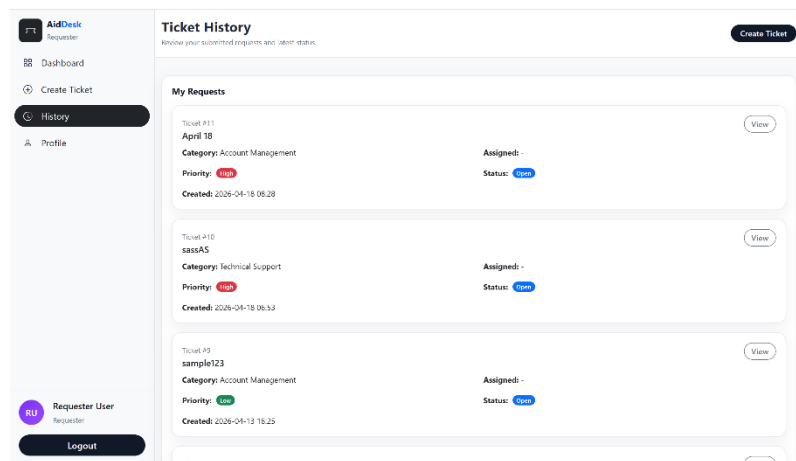


Figure 19: Requester_history

12. Development Tools

- Laravel
- Blade templating engine
- Eloquent ORM
- MySQL
- Bootstrap
- Laravel migratio

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